

PREFACE

Within the nuclear industry, there has been a strong demand for realistic analysis tools based on best-estimate modeling for application in the thermal hydraulic analyses of nuclear reactor systems. This demand has led to the development of such advanced codes as RELAP5, COBRA-TF, TRAC, CATHARE, etc.

Korea Advanced Energy Research Institute (KAERI) conceived and started the development of MARS code with the main objective of producing a state-of-the-art realistic thermal hydraulic systems analysis code with multi-dimensional analysis capability. MARS achieves this objective by very tightly integrating the one dimensional RELAP5/MOD3 with the multi-dimensional COBRA-TF codes.

RELAP5 is a versatile and robust code based on one-dimensional two-fluid formulation, and the code includes many generic component and special process models. COBRA-TF is based on three-dimensional two-fluid, three-field formulation and was developed especially for the reflood thermal-hydraulics of large-break Loss of Coolant Accident (LOCA).

The method of integration of the two codes is based on the dynamic link library techniques, and the system pressure equation matrices of both codes are implicitly integrated and solved simultaneously. In addition, the Equation-Of-State (EOS) for the light water was unified by replacing the EOS of COBRA-TF by that of the RELAP5.

Other developmental objectives are to modernize the coding and to provide some user convenience by providing simple but effective Graphic User Interface (GUI). With some key programming techniques dating back to 1960's, the coding of RELAP5 lags behind the current level of the computer industry. By taking advantages of structured programming techniques provided in modern compiler, FORTRAN 90, the RELAP part of the MARS code was thoroughly re-structured from the original RELAP coding with the objectives of improving the legibility, maintenance and ease of modification. The input scheme was unified whereby the original formatted input scheme of the COBRA-TF has been replaced with the free-format scheme of the RELAP. By this unification, a single free format input file is used for the two parts, one-dimensional and multi-dimensional. Some simple GUI are provided for the program execution and for the on-line showing of the calculation results. The MARS code has been successfully developed and it is being used in many key areas of the nuclear industry.

This input manual provides a complete list of input required to run MARS. The manual is divided largely into two parts, namely, the one-dimensional part and the multi-dimensional part. The inputs for auxiliary parts such as minor edit requests and graph formatting inputs are shared by the two parts and as such mixed input is possible. The overall structure of the input is modeled on the structure of the RELAP5 and as such the layout of the manual is very similar to that of the RELAP. This similitude to RELAP5 input is intentional as this input scheme will allow minimum modification between the inputs of RELAP5 and MARS. MARS development team would like to express its appreciation to the RELAP5 Development Team and the USNRC for making this manual possible.